Efficient Frontier Shift: Pre- vs. Post-COVID Indian Equities

IQF Midterm Project

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Research Question

Does the efficient frontier for Indian equities shift meaningfully between **pre-COVID (2015–2020)** and **post-COVID (2021–2024)**, and what structural forces drive the shift?

Why it matters

- COVID was a sharp structural break in global markets
- India's equity market saw unprecedented post-pandemic stimulus and sectoral rotation
- A shift in the frontier implies a change in the *fundamental* risk-return trade-off available to investors

Dataset

Universe

- 45 stocks from NIFTY 100 (as of Jan 1, 2015)
- Fixed universe $\rightarrow$ clean pre/post comparison
- Source: Yahoo Finance `.NS` tickers

Periods (Option C)

- Pre-COVID: Jan 2015 – Jan 2020
- Crisis window excluded: Feb 2020 – Jun 2021
- Post-COVID: Jul 2021 – Dec 2024
- Analyzed at daily & weekly frequency

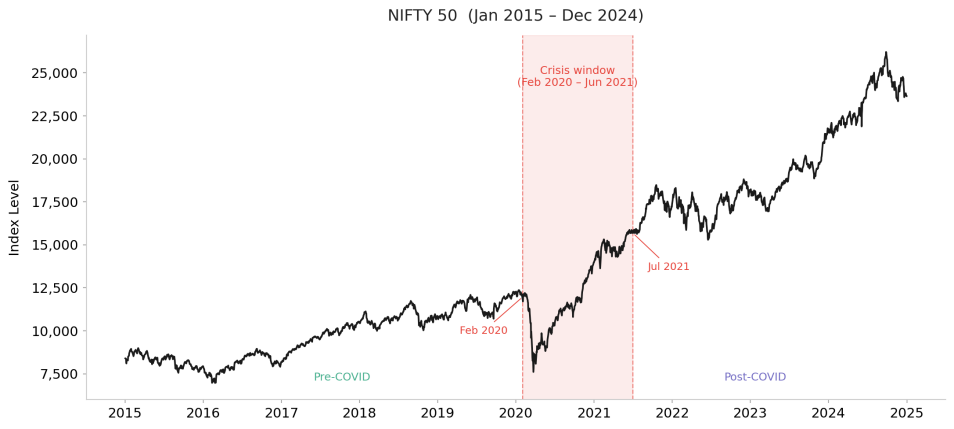
Estimation

- Log returns; annualized ($\times 252$ daily, $\times 52$ weekly)
- Ledoit-Wolf shrinkage covariance (accounts for finite-sample noise)
- Risk-free rate: RBI 91-day T-Bill yield (pre: 6.54%, post: 5.78%)

Data quality

- Survivorship bias acknowledged: fixed universe biases upward
- Delisted/merged tickers retain Yahoo historical data

NIFTY 50 Index: 2015–2024



- NIFTY fell $\approx$26% peak-to-trough (Feb–Mar 2020); the crisis window (shaded) is *excluded* from both estimation periods
- Post-COVID window captures a clean bull-market regime, free of crash-induced distortions

Methodology

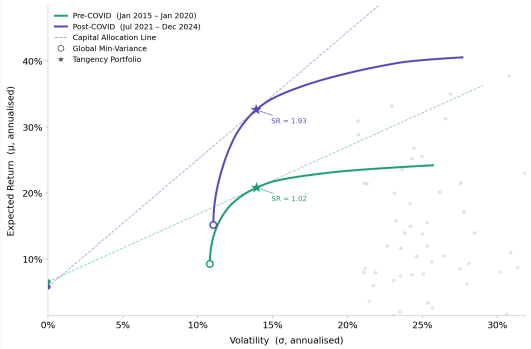
1. **Compute statistics** — log returns annualized; Ledoit-Wolf shrinkage applied to $\mathbf{\Sigma}$ for each period
2. **Plot individual assets** — each stock's (σ_i, μ_i) shown as a reference scatter; no simulated feasible set (avoids concentration-of-measure issues at $N = 45$)
3. **Trace the efficient frontier** — solve per-point SLSQP optimization

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \mathbf{\Sigma}\, \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu^*, \ \mathbf{w}^\top \mathbf{1} = 1, \ \mathbf{w} \geq 0$$

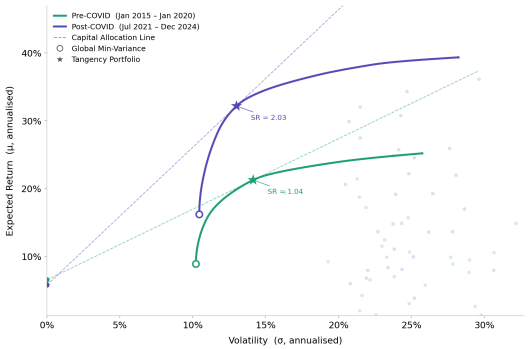
   sweeping μ^* from GMV to the highest-return asset
4. **Tangency portfolio** — maximize Sharpe ratio with RBI T-bill as r_f; compare pre vs. post weights

Efficient Frontier — Daily & Weekly

Daily (T/N ≈ 28 pre, 19 post)

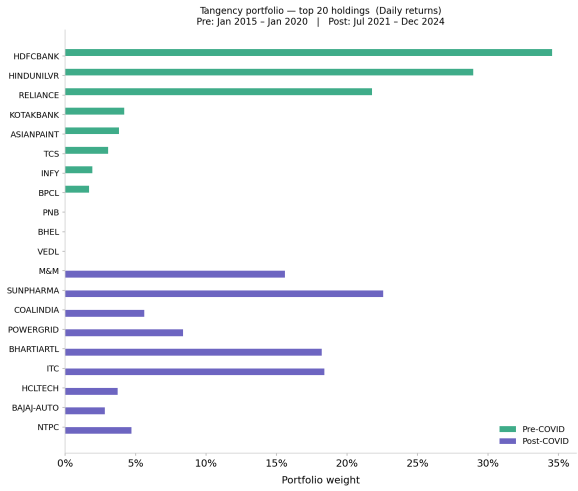


Weekly (T/N ≈ 6 pre, 4 post)



The post-COVID frontier lies *above and to the right* in both cases — the risk-return trade-off improved materially.

Tangency Portfolio: Sectoral Rotation



Pre-COVID top holdings

- HDFCBANK 34.5%
- HINDUNILVR 29.0%
- RELIANCE 21.8%

Post-COVID top holdings

- SUNPHARMA 22.6%
- ITC 18.4%
- BHARTIARTL 18.2%
- M&M 15.6%

Complete rotation out of **financials & FMCG** into **pharma, telecom, energy, autos** — the optimizer's view of where

Interpreting the Shift

Why did the frontier shift northeastward?

Demand-side drivers

- Post-COVID fiscal stimulus and infra spending (PLI schemes, BHEL, TATAPOWER)
- Domestic retail investor surge: Demat accounts doubled 2020–2022
- Foreign capital inflows as India became a China+1 alternative

Structural / sectoral drivers

- Cyclicals and metals outperformed on commodity supercycle (VEDL, HINDALCO, TATASTEEL)
- Digital transformation boosted IT returns (HCL, TCS)
- Pharma re-rated as strategic post-COVID (SUNPHARMA)
- Financials drag: regulatory tightening + HDFC-HDFCBANK merger uncertainty

Stock-Level: Biggest Winners

Ticker	Company	Pre	Post	Δ
VEDL	Vedanta	+8.0%	+51.2%	+43.2 pp
TATAPOWER	Tata Power	−4.9%	+48.7%	+53.5 pp
BHEL	Bharat Heavy Electricals	−20.7%	+46.8%	+67.5 pp
M&M	Mahindra & Mahindra	+0.4%	+43.0%	+42.6 pp
SUNPHARMA	Sun Pharmaceutical	−10.4%	+34.1%	+44.5 pp
HCLTECH	HCL Technologies	+8.6%	+32.4%	+23.8 pp

Industrials, metals, pharma, and IT led the post-COVID recovery.
BHEL and TATAPOWER flipped from worst to top performers — a full sectoral reversal.

Stock-Level: Pre-COVID Stars That Faded

Ticker	Company	Pre	Post	Δ
KOTAKBANK	Kotak Mahindra Bank	+21.1%	+4.6%	−16.5 pp
HDFCBANK	HDFC Bank	+19.7%	+10.6%	−9.1 pp
HINDUNILVR	Hindustan Unilever	+17.9%	+7.9%	−10.0 pp
ASIANPAINT	Asian Paints	+17.8%	+9.4%	−8.4 pp
BPCL	BPCL	+23.5%	+15.3%	−8.2 pp

Financials and consumer staples — the pre-COVID defensives — underperformed significantly post-COVID.

These are also the stocks that *dominated the pre-COVID tangency portfolio*.

Robustness Across Frequencies

Frequency	Pre-COVID		Post-COVID	
	Return	Vol.	Return	Vol.
Daily (T/N $\approx$ 28 / 19)	7.91%	28.95%	24.46%	33.17%
Weekly (T/N $\approx$ 6 / 4)	7.42%	27.95%	23.73%	32.58%
Avg. Δ Return		**+16.4 pp**		
Avg. Δ Vol.		+4.9 pp		

The $\approx$16 pp return uplift is consistent across both frequencies.
Weekly has lower T/N; Ledoit-Wolf shrinkage is higher (0.06–0.07) but the qualitative result holds.

Limitations

- **Survivorship bias** — fixed 2015 universe excludes stocks that failed or were delisted; biases results upward
- **Sample depth** — daily T/N $\approx$ 19–28 (reliable); weekly T/N $\approx$ 4–6 (higher shrinkage applied, 0.06–0.07); results should be interpreted with this in mind
- **Long-only constraint** — rules out short selling; actual attainable frontier may be wider
- **Transaction costs & liquidity** — not modeled; tangency portfolios are concentrated and may be costly to implement at scale
- **Causal attribution** — the drivers discussed are correlational; isolating macro causes requires further regression analysis

Future Roadmap

1. **Statistical significance test** — bootstrap or permutation test for the frontier shift; quantify the probability that the observed gap is due to sampling noise
2. **Factor attribution** — decompose the tangency weight rotation into sector and style factors (value, momentum, quality); link the structural shift to identifiable risk premia
3. **Transaction cost modeling** — estimate rebalancing costs for the tangency portfolio; assess whether the Sharpe improvement survives after costs
4. **Out-of-sample validation** — apply post-COVID tangency weights to 2025 data; test whether the frontier shift was a durable regime change or a bull-market artifact

Conclusions

1. The efficient frontier for NIFTY 100 equities **shifted decisively upward and to the right** post-COVID
2. The return uplift ($\approx +16$ pp annualized) **vastly outpaced the volatility increase** ($\approx +5$ pp), improving the risk-adjusted opportunity set
3. The shift is **robust across daily and weekly** observation frequencies, with consistent Sharpe improvements
4. The tangency portfolio underwent a **complete sectoral rotation** — financials and FMCG replaced entirely by pharma, telecom, energy, and autos
5. COVID was not merely a transient shock; it **structurally altered the investment landscape** for Indian equities

Thank you

Questions?